Quantification: I534_z11c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	39
interior lacuna count	31
frame-edge lacuna count	8

Interior lacunae only: mean, median and sample SD (n = 31)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2372.23	1679	2532.03	px^2
major_axis_length_px	lacuna shape	74.906	72.0848	33.9434	px
minor_axis_length_px	lacuna shape	39.9874	35.4441	21.9689	px
aspect_ratio	lacuna shape	2.0205	1.9891	0.6037	unitless
eccentricity	lacuna shape	0.8085	0.8644	0.1667	unitless
solidity	lacuna shape	0.7438	0.724	0.0938	unitless
roots_count	headline	9.9677	9	5.5527	count
ring_length_r30_px	headline	271.161	240	129.006	px
ring_length_r60_px	reported	498.807	432	261.211	px
owned_length_px	ownership-dependent	562.253	398.392	456.242	px
edge_count	ownership-dependent	28.1935	23	21.4933	count
mean_edge_length_px	ownership-dependent	19.791	19.1684	5.6484	px
perimeter_px	normalised	244.171	233.38	126.651	px
ring_area_r30_px2	normalised	8121.42	6884	3561.33	px^2
ring_area_r60_px2	normalised	15763.5	13291	7270.42	px^2
in_frame_fraction_r30	normalised	0.9891	1	0.0298	unitless
in_frame_fraction_r60	normalised	0.9668	1	0.0709	unitless
ring_density_r30	normalised	0.0345274	0.0345449	0.0108847	px^-1
ring_density_r60	normalised	0.0325079	0.0346367	0.0102082	px^-1
roots_per_100px_perimeter	normalised	4.2202	4.137	1.3818	per 100 px
ring_attached_length_r30_px	network v2	246.161	199	131.231	px
ring_attached_length_r60_px	network v2	396.903	314	261.287	px
ring_length_w_r30_px	network v2	307.76	275.522	146.001	px
ring_length_w_r60_px	network v2	566.181	494.931	297.06	px
sholl_crossings_r10	network v2	5.9677	5	3.2607	count
sholl_crossings_r20	network v2	6.871	6	3.7483	count
sholl_crossings_r30	network v2	7.2903	7	4.1085	count
loop_count	loops, report only	1.3548	0	2.3881	count
loop_edge_share	loops, report only	0.1031	0	0.1445	share
ring_width_mean_r30_px	width	7.7798	7.6247	1.1681	px

Per field
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measure	value	units
analysed_area_px2	884210	px^2
total_skeleton_length_px	25118	px
canalicular_length_density_per_px	0.0284073	px^-1
junction_count	547	count
junction_density_per_px2	0.00061863	px^-2
lacuna_count	39	count

lacunae_per_px2	4.411e-05	px^-2
skeleton_component_count	316	count
mean_component_length_px	79.4873	px
median_component_length_px	23	px
field_length_density_w_per_px	0.0322136	px^-1
field_density_without_flagged_per_px	0.0298285	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	197	px
unattached_component_count	13	count
canalicular_length_density_attached_per_px	0.0284073	px^-1
field_length_density_with_unattached_per_px	0.0286301	px^-1
loop_count	55	count
loop_edge_share	0.1394	share
edge_count	1392	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.2143	px
width_median_px	8.2462	px
width_p10_px	4	px
width_p90_px	11.6619	px
width_pixels_used	23536	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	5018	113.612	66.0072	1.7212	0.8139	0.7134	-1.5671	33.3908	243.731	True
2	1429	75.9455	34.6321	2.1929	0.89	0.5788	0.9821	47.7124	162.336	False
3	468	39.8411	18.6563	2.1355	0.8836	0.8	-1.37	7.2714	740.387	True
4	12478	404.05	54.0339	7.4777	0.991	0.6138	1.5354	38.8529	844.287	True
5	1943	78.2261	36.7629	2.1279	0.8827	0.7692	-1.4856	56.1004	636.36	False
6	1748	63.7004	52.4126	1.2154	0.5683	0.625	1.3536	36.7677	399.808	False
7	482	37.8235	20.2932	1.8638	0.8439	0.7226	-0.6658	78.6515	500.948	False
8	2001	74.781	36.9486	2.0239	0.8694	0.794	-1.1924	101.66	153.378	False
9	499	44.514	18.7039	2.3799	0.9074	0.7607	1.3732	93.7014	712.353	False
10	4592	90.8676	67.3599	1.349	0.6712	0.8423	-1.2953	137.617	765.059	False
11	902	70.1443	23.2471	3.0173	0.9435	0.6677	-1.2472	150.552	535.413	False
12	18263	251.192	114.111	2.2013	0.8909	0.7282	-1.0667	218.389	97.7195	True
13	2333	67.9441	45.1639	1.5044	0.7471	0.9554	-0.4874	234.38	1000.91	True
14	11887	168.982	95.145	1.7761	0.8264	0.8606	-1.2985	286.803	681.158	False
15	697	50.8222	20.9969	2.4205	0.9107	0.7383	1.293	287.073	212.017	False
16	3532	87.4182	55.1855	1.5841	0.7756	0.7638	1.235	329.842	344.026	False
17	4129	107.816	50.0459	2.1543	0.8857	0.9077	-1.34	378.661	498.241	False
18	889	60.4502	27.413	2.2052	0.8913	0.6247	-1.2388	363.547	49.8526	False
19	514	40.4388	21.1806	1.9092	0.8519	0.6993	0.9864	406.87	807.179	False
20	587	53.646	23.5495	2.278	0.8985	0.6501	-0.399	433.826	208.523	False
21	1735	68.9793	37.2098	1.8538	0.842	0.6893	-0.903	463.884	594.474	False
22	751	42.1527	25.0435	1.6832	0.8044	0.8632	-1.1817	459.581	482.401	False
23	508	31.1659	30.9736	1.0062	0.1109	0.6737	0.679	475.947	802.87	False
24	5417	114.374	70.6956	1.6178	0.7861	0.6168	1.316	527.301	55.7192	True

25	3037	99.839	58.2931	1.7127	0.8118	0.571	1.4007	558.019	301.395	False
26	4073	115.2	49.8465	2.3111	0.9015	0.7391	0.3504	603.966	867.145	False
27	2337	72.0848	52.1015	1.3835	0.6911	0.6974	-1.4357	607.677	651.435	False
28	1679	74.0209	37.2141	1.9891	0.8644	0.724	-0.9638	634.934	701.784	False
29	7591	130.597	95.8386	1.3627	0.6793	0.6789	-0.2363	683.477	82.1523	False
30	4140	142.999	43.9592	3.253	0.9516	0.7236	-1.1238	670.561	384.423	False
31	554	30.7716	24.9383	1.2339	0.5858	0.895	1.0673	729.574	697.603	False
32	549	48.7614	15.8873	3.0692	0.9454	0.8356	-0.9101	738.328	786.444	False
33	640	57.661	17.8443	3.2313	0.9509	0.7018	-0.3969	793.93	358.258	False
34	2036	78.7104	35.4441	2.2207	0.8929	0.8879	-1.002	828.682	94.3777	False
35	45573	496.877	154.577	3.2144	0.9504	0.7075	1.5056	966.648	733.764	True
36	1332	74.3708	26.5147	2.8049	0.9343	0.75	-1.2358	829.074	388.601	False
37	506	33.5192	20.2873	1.6522	0.796	0.8988	-0.9746	842.302	981.741	False
38	6240	115.676	84.3355	1.3716	0.6844	0.7226	-1.0938	881.946	274.251	False
39	1277	51.634	38.2716	1.3491	0.6713	0.7854	1.2894	872.15	1003.8	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	15	337	427	568.174	29	19.5922	6.7223
2	12	274	358	364.877	23	15.8642	6.4715
3	1	19	19	23.8995	2	11.9497	4.0262
4	23	408	523	465.274	26	17.8952	6.6768
5	10	267	467	398.392	26	15.3228	7.168
6	9	352	485	357.208	22	16.2367	7.3539
7	11	264	366	438.534	27	16.242	7.5091
8	7	240	321	161.439	10	16.1439	6.8457
9	7	133	157	113.225	8	14.1532	6.1928
10	14	382	549	632.558	33	19.1684	7.339
11	8	315	698	776.683	38	20.439	7.6403
12	18	515	788	919.666	48	19.1597	7.5529
13	6	155	358	401.421	19	21.1274	8.7355
14	26	659	1175	1353.12	67	20.1958	8.1509
15	7	204	384	431.333	23	18.7536	8.7157
16	16	500	1052	1999	93	21.4946	8.5149
17	11	341	658	1054.25	55	19.1681	8.3777
18	7	179	235	246.492	10	24.6492	6.1524
19	7	218	484	1118.03	52	21.5006	7.5444
20	6	230	501	349.907	23	15.2133	8.3811
21	12	360	668	1001.49	35	28.614	7.7108
22	5	155	258	175.125	10	17.5125	8.2917
23	6	248	432	392.049	18	21.7805	6.9808
24	19	351	508	724.855	34	21.3193	5.9354
25	19	379	844	1380.11	69	20.0016	7.8666
26	15	420	955	815.21	40	20.3803	7.0277
27	15	277	585	667.245	36	18.5346	7.6247
28	9	167	339	207.48	7	29.64	7.1073
29	23	499	869	903.436	46	19.6399	7.5211
30	9	207	349	267.61	17	15.7418	7.6147
31	5	168	322	118.498	6	19.7496	8.2887
32	4	177	394	201.723	5	40.3446	8.1657
33	1	35	62	7.8284	1	7.8284	12.9591
34	8	232	483	506.031	22	23.0014	8.1991
35	39	777	1166	1686.33	88	19.1629	8.6448
36	6	119	213	157.468	10	15.7468	7.2164
37	4	178	381	509.877	27	18.8843	8.1717
38	10	227	419	323.622	15	21.5748	8.0692
39	3	95	118	127.569	8	15.9461	

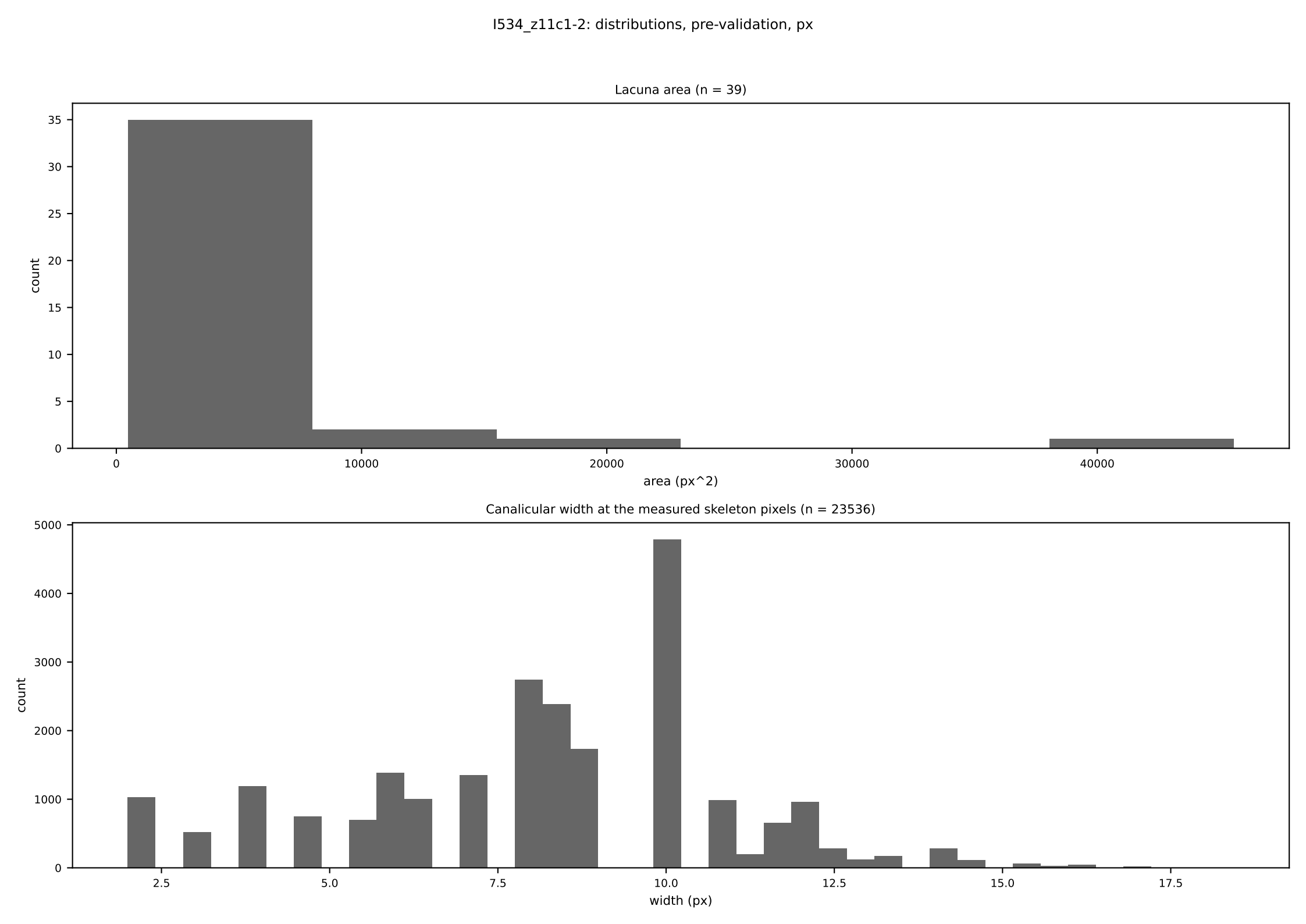
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	494.351	8231	11715	0.7421	0.6708	0.0409428	0.036449	3.0343
2	282.291	4862	7142	0.9455	0.8134	0.0563554	0.050126	4.2509
3	124.669	647	647	0.5879	0.5566	0.0293663	0.0293663	0.8021
4	1255.09	17088	22292	0.693	0.5601	0.0238764	0.0234613	1.8325
5	253.279	6884	10939	1	0.9073	0.0387856	0.0426913	3.9482
6	270.635	8155	13291	0.8695	0.7737	0.0431637	0.0364909	3.3255
7	123.054	5704	10031	1	1	0.0462833	0.0364869	8.9392
8	236.936	5018	6853	1	1	0.0478278	0.0468408	2.9544
9	120.711	2887	3931	1	1	0.0460686	0.039939	5.799
10	341.505	9524	15294	1	1	0.0401092	0.0358964	4.0995
11	176.167	7534	16220	1	1	0.0418105	0.0430333	4.5412
12	1221.79	20165	28289	0.8577	0.7956	0.0255393	0.0278554	1.4732
13	199.782	5730	14247	0.6804	0.6335	0.0270506	0.0251281	3.0033
14	548.257	17191	32458	1	1	0.038334	0.0362006	4.7423
15	136.125	6658	12133	1	1	0.0306398	0.0316492	5.1423
16	297.706	11030	23116	1	1	0.0453309	0.0455096	5.3744
17	305.605	10599	17028	1	1	0.0321729	0.0386422	3.5994
18	191.167	7042	12138	0.9189	0.7962	0.0254189	0.0193607	3.6617
19	132.267	5680	13004	1	1	0.0383803	0.0372193	5.2923
20	139.267	6658	17676	1	1	0.0345449	0.0283435	4.3083
21	238.451	9266	19320	1	1	0.0388517	0.0345756	5.0325
22	122.953	5564	11136	1	1	0.0278577	0.0231681	4.0666
23	122.953	5620	11532	1	1	0.0441281	0.037461	4.8799
24	490.399	11727	20264	0.8061	0.7376	0.0299309	0.0250691	3.8744
25	378.475	12733	26807	1	1	0.0297652	0.0314843	5.0201
26	373.546	12922	27828	1	1	0.0325027	0.0343179	4.0156
27	290.877	7526	16187	1	1	0.0368057	0.0361401	5.1568
28	233.38	6410	12080	1	1	0.026053	0.0280629	3.8564
29	555.955	15261	25089	0.9401	0.8614	0.0326977	0.0346367	4.137
30	383.889	12892	25468	1	1	0.0160565	0.0137035	2.3444
31	97.9828	5438	10968	1	1	0.0308937	0.0293581	5.1029
32	115.539	5958	13707	1	1	0.029708	0.0287444	3.462
33	139.054	4710	7157	1	1	0.007431	0.00866285	0.7191
34	222.693	8786	20073	1	0.9917	0.0264056	0.0240622	3.5924
35	2556.07	33726	52683	0.6475	0.572	0.0230386	0.0221324	1.5258
36	196.066	6012	12715	1	1	0.0197937	0.0167519	3.0602
37	96.0833	3593	8752	0.9896	0.8284	0.0495408	0.0435329	4.1631
38	446.416	13647	28597	1	1	0.0166337	0.0146519	2.2401
39	206.752	3172	3789	0.6755	0.6216	0.0299496	0.0311428	1.451

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	334	421	381.806	480.818	9	8	4
2	271	315	309.35	406.605	9	6	3
3	16	16	20.8995	20.8995	0	0	0
4	387	446	454.304	589.5	12	8	4
5	191	272	304.764	530.789	6	5	9
6	348	425	392.836	534.433	6	9	5
7	253	314	295.794	412.877	5	5	5
8	151	151	275.522	367.291	5	4	4
9	108	108	155.024	180.51	2	3	1
10	363	516	438.818	626.772	9	10	9

11	278	600	361.149	803.311	6	10	9
12	465	540	581.902	895.394	12	14	9
13	140	252	165.669	393.836	3	4	4
14	639	1127	726.487	1314.56	15	15	17
15	196	285	235.38	437.676	5	6	4
16	486	1022	568.688	1196.76	10	16	18
17	327	612	395.505	761.553	8	9	12
18	153	156	207.794	271.321	5	2	3
19	184	413	243.924	553.274	5	8	6
20	180	298	267.865	576.529	3	6	8
21	340	593	412.605	776.039	6	7	9
22	145	178	175.297	287.551	4	4	4
23	221	303	279.966	494.931	5	5	8
24	327	352	403.576	592.541	7	3	5
25	370	705	430.818	954.365	11	10	10
26	408	696	481.203	1073.34	8	12	12
27	232	384	316.179	665.186	3	6	7
28	153	183	181.853	382.019	6	3	3
29	451	748	578.399	1005.9	13	14	13
30	199	281	226.853	381.007	5	4	7
31	149	248	184.054	363.978	4	6	4
32	115	165	202.995	451.789	4	7	7
33	10	10	37.799	67.6985	0	1	2
34	232	429	267.794	549.902	6	6	7
35	716	1035	866.323	1308.97	31	16	9
36	119	159	135.397	236.238	2	2	3
37	178	356	199.297	424.836	2	5	8
38	181	252	251.137	462.576	7	7	9
39	95	101	104.498	132.297	4	2	1


I534_z11c1-2: distributions, pre-validation, px
Lacuna area (n = 39)
count
area (px^2)
0
5
10
15
20
25
30
35
0
10000
20000
30000
40000
Canalicular width at the measured skeleton pixels (n = 23536)
count
width (px)
0
1000
2000
3000
4000
5000
2.5
5.0
7.5
10.0
12.5
15.0
17.5